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**Do Machine Learning Models Improve Stock Return
Prediction?**

Evidence from S&P 500 Constituent Markets and Dimension Sensitivity (2015–2024)

Relatrice

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Abstract

Can machine learning models pick US stocks better than traditional financial formulas? I test this question using 10 years of daily S&P 500 stock data from 2015 to 2024. My dataset tracks 59 key features for each stock. These include price momentum, company financial ratios, interest rates, market volatility, news sentiment, and risk factor betas.

To make sure models do not use future information by mistake, I test them using a strict 5-year rolling timeline from 2020 to 2024. The results are clear. Deep learning models beat standard linear models (like Fama-MacBeth regressions or LASSO) and decision trees (like Random Forest and XGBoost). My custom model, called **TFDMGA**, performs the best. It achieves an out-of-sample daily rank correlation (*IC*) of +0.0348, an Information Ratio (*ICIR*) of 3.12, a ROC AUC of 0.6120, and a directional accuracy of 56.84%. A formal Diebold-Mariano test confirms that TFDMGA significantly outperforms standard PyTorch LSTM models ($DM = 2.41, p = 0.016$).

Next, I simulate real trading with a 1-day execution delay and 10 bps trading fees. A \$1,000 USD investment starting in 2020 grows to \$3,120.99 USD using a standard LSTM strategy. Adding a 2:1 Take-Profit and Stop-Loss rule boosts capital to \$4,368.50 USD for LSTM and \$6,482.10 USD for TFDMGA. However, when I run Fama-French 5-factor regressions on the returns, the net strategy alpha is zero ($\hat{\alpha} = -0.18\%, p = 0.976$). This shows that the extra returns come from managing systematic market risk factors dynamically, not from finding free money in the market.

Keywords: Machine Learning, Stock Return Prediction, Deep Learning, Transformer Attention, Fama-French Models, Transaction Costs, Trading Strategies.

يا رب زدني علما

“And say, ‘My Lord, increase me in knowledge.’ ”

, The Holy Qur’an, Surah Ta-Ha (20:114)

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Contents

Abstract	i
Dedication	ii
Acknowledgements	iii
1 Introduction	1
1.1 Background and Economic Motivation	1
1.2 Research Questions and Core Hypotheses	2
1.3 Summary of Key Empirical Findings	3
1.4 Structure of the Thesis Manuscript	3
2 Literature Review	5
2.1 Introduction	5
2.2 Theoretical Asset Pricing Foundations	6
2.2.1 Equilibrium Foundations: The CAPM and Arbitrage Pricing Theory	6
2.2.2 The Fama-MacBeth Two-Pass Regression Framework	7
2.2.3 Multi-Factor Models: Fama-French, Carhart, and Pastor- Stambaugh	8
2.3 The Factor Zoo Controversy, Multiple Testing, and Shrinkage	9
2.3.1 The Proliferation of Anomalies and Cochrane's "Factor Zoo"	9
2.3.2 Multiple Testing Inflation and False Discovery Rates	10

2.3.3	High-Dimensional Regularized SDF Estimation	10
2.4	High-Dimensional SDF Estimation & Characteristic Interactions	11
2.4.1	Non-Parametric Characteristic Pruning via Group LASSO	11
2.4.2	Sparse Signals and Latent Factor Shrinkage	12
2.5	Machine Learning and Deep Learning Paradigms	13
2.5.1	Empirical Asset Pricing Benchmark: Gu, Kelly, and Xiu (2020)	13
2.5.2	Characteristics as Dynamic Risk Exposures: IPCA	14
2.5.3	Deep Learning and Non-Parametric SDF Estimation	14
2.5.4	Economic Frictions and Limits to Arbitrage	15
2.6	Financial Time-Series Econometrics & Inference	15
2.6.1	Heteroskedasticity and Autocorrelation Consistent (HAC) Infer- ence	15
2.6.2	Out-of-Sample Predictive Accuracy: Diebold-Mariano and HLN Corrections	16
2.6.3	Data Mining Diagnostics: Reality Check and Superior Predictive Ability	17
2.7	Financial ML Protocols & Overfitting Mitigation	17
2.7.1	Methodological Pitfalls in Financial Machine Learning	17
2.7.2	Validation Protocols	18
2.7.3	Microstructure Frictions & Transaction Cost Taxonomy	18
2.8	Synthesis of Core Literature Benchmarks	18
2.9	Summary	19
3	Data & Methodology	20
3.1	Data Pipeline and Sample Construction	20

3.1.1	Point-in-Time Bloomberg Filing Date Alignment	20
3.1.2	Survivorship-Bias-Free Dynamic Panel	21
3.2	Feature Space Taxonomy and Selection	21
3.2.1	Economic Rationale for Feature Set Size (59 Variables vs. 120+ Zoo Features)	21
3.3	Feature Normalization and Selection Protocol	22
3.4	Baseline Econometric & Machine Learning Formulations	24
3.4.1	Fama-MacBeth 2-Pass Linear Regression & Newey-West HAC Estimation	25
3.4.2	Regularized Linear Models: ElasticNet, LASSO, & Ridge	29
3.4.3	Non-Linear Tree Ensembles: Random Forest & XGBoost	32
3.4.4	Baseline Sequence Encoder: PyTorch LSTM	36
3.5	Custom Architecture: TFDMGA Network	37
3.5.1	Temporal Feature Encoding via Causal 1D Dilated TCN	38
3.5.2	Sequential Directed Ring Attention Cascade	39
3.5.3	3-Way Macro Dynamic Gating Network	40
3.5.4	Residual Fusion Block & Pre-LN Transformer Encoder	42
3.5.5	Multi-Task Multi-Head Objective Function	43
3.6	Cross-Validation & Walk-Forward Evaluation Scheme	44
3.7	Hyperparameter Search Space & Training Setup	45
3.8	Computational Complexity and Resource Profile	46
4	Empirical Results & Backtesting	47
4.1	Introduction	47
4.2	Contemporaneous Fama-MacBeth OLS Baseline	47

4.3	Mathematical Derivation & Empirical Reconciliation of LASSO Shrinkage Collapse	48
4.4	SHAP Feature Interpretability Analysis	49
4.5	Out-of-Sample Predictive Evaluation	50
4.6	Ablation Study of Neural Architecture Components	51
4.7	Robustness Checks and Sensitivity Analysis	52
4.8	Portfolio Construction & Execution Friction	52
4.9	Transaction Cost Sensitivity & \$1,000 USD Account Compounding . . .	52
4.10	Fama-French 5-Factor Spanning Regressions	54
5	Conclusion & Future Extensions	56
5.1	Summary of Research Findings	56
5.2	Theoretical and Practical Implications	57
5.3	Limitations of the Study	58
5.4	Future Research Extensions	58
	Bibliography	60

List of Figures

3.1	Bloomberg Point-in-Time Data Pipeline and Staleness Truncation Flowchart	21
3.2	Variable Taxonomy and Feature Group Modality Split (53 ML Inputs + 6 Factor Betas = 59 Total)	22
3.3	Spearman Rank Correlation Heatmap of Selected 53 ML Features	23
3.4	Stage 3 In-Sample Permutation Feature Importance Rankings across Modalities	24
3.5	Architecture Diagram of the Temporal Fusion Deep Multimodal Gated Attention (TFDMGA) Network	37
3.6	5-Fold Expanding-Window Walk-Forward Validation Protocol (2015–2024)	44
4.1	Global SHAP Mean Absolute Feature Importance Rankings (21-Day Horizon)	49
4.2	SHAP Summary Beeswarm Plot Illustrating Feature Value Impact on Model Predictions	50
4.3	Quintile Portfolio Sorting, 1-Day Execution Delay, and Turnover Rebalancing Flowchart	52
4.4	Out-of-Sample Cumulative Return Trajectories (Net 10 bps, 2020–2024 Folds)	53
4.5	Rolling 126-Day Annualized Sharpe Ratios across Model Architectures	53
4.6	Impact of 2:1 Take-Profit/Stop-Loss Risk Overlay (+4% / -2%) on Portfolio Returns	54

4.7	Maximum Drawdown Mitigation During Macro Stress Periods (COVID 2020 & Fed Hikes 2022)	54
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List of Tables

2.1	Comprehensive Synthesis of Core Literature Benchmarks	19
3.1	Hyperparameter Search Space and Optimal Tuned Parameters Across Models	45
3.2	Computational Complexity, Parameter Count, and Resource Execution Profile	46
4.1	Fama-MacBeth Daily Cross-Sectional Regression Risk Premia (2015– 2024)	47
4.2	Out-of-Sample Predictive Performance Metrics (2020–2024 Test Folds – Master Panel 53 Features)	50
4.3	Systematic Component Ablation Study of the TFDMGA Network . . .	51
4.4	Predictive Accuracy and Account Compounding Across Prediction Hori- zons and Fees	52
4.5	\$1,000 USD Account Growth Under Transaction Cost Scenarios (2020– 2024)	53
4.6	Fama-French 5-Factor Spanning Regressions with Newey-West HAC Errors	55

1. Introduction

1.1 Background and Economic Motivation

Why do stock prices move differently from one another, and can we predict these movements? Economists have tried to answer this question for over 50 years [1, 2]. Classical financial models assume that stock returns depend linearly on a few static risk factors. These include the Capital Asset Pricing Model (CAPM) by Sharpe [3] and Lintner [4], the Arbitrage Pricing Theory by Ross [1976], and the 5-factor model by Fama and French [2015].

Real stock markets are much more complex. Stock prices show sudden regime shifts, fat-tailed risks, and non-linear patterns. For example, valuation metrics like Price-to-Earnings ratios behave differently during market crashes than during bull runs [7, 8]. Standard linear regressions struggle when tracking dozens of stock characteristics at once, leading to overfitting or missed signals.

Machine learning offers a flexible way to capture these complex interactions [7–9]. Models like decision trees [10, 11], neural networks [12], and Transformers [13] can learn patterns directly from financial data.

However, stock data is very noisy. Naive machine learning models often fail because they accidentally use future data, overfit historical noise, or ignore trading fees [14–16]. In this thesis, I test whether machine learning models can reliably predict stock returns when tested under strict, real-world trading rules across S&P 500 stocks from 2015 to 2024.

1.2 Research Questions and Core Hypotheses

I organize my research around four central questions:

Research Question 1 (Predictability): Do deep learning models (LSTM, TFDMGA) predict stock returns better than standard linear regressions (OLS, LASSO) and decision trees (Random Forest, XGBoost)?

Hypothesis 1: Deep sequence models outperform linear models because they track historical time trends and non-linear feature interactions directly.

Research Question 2 (Multimodal Data Integration): Does combining price technicals, financial ratios, macro data, and news sentiment improve forecast stability?

Hypothesis 2: Combining multiple data types allows attention mechanisms to reweight features as market conditions change, improving signal stability.

Research Question 3 (Trading Profits & Costs): Can these model predictions build profitable trading strategies after deducting 10 bps trading fees and 1-day execution delays?

Hypothesis 3: Even though daily rebalancing incurs fee drag, pairing deep learning models with a 2:1 Take-Profit/Stop-Loss rule protects capital and compounds returns.

Research Question 4 (Market Alpha vs. Factor Exposure): Are the extra trading returns coming from unpriced market alpha ($\alpha > 0$) or from systematic factor risks ($\alpha = 0$)?

Hypothesis 4: The strategy returns are fully explained by standard Fama-French risk factors, showing that machine learning models act as dynamic factor managers rather than finding free money.

1.3 Summary of Key Empirical Findings

Using a 5-year rolling test timeline from 2020 to 2024, my key empirical findings are:

1. **Top Forecasting Model:** My custom **TFDMGA** neural network achieves the highest predictive accuracy across all models. It records an out-of-sample daily rank correlation of $IC = +0.0348$, an Information Ratio of $ICIR = 3.12$, a ROC AUC of 0.6120, and a directional accuracy of 56.84%.
2. **Statistical Comparison:** A formal Diebold and Mariano [17] test confirms that TFDMGA significantly outperforms a standard PyTorch LSTM model ($DM = 2.41, p = 0.016$).
3. **Account Compounding:** Under 10 bps trading fees and 1-day execution delays, an initial \$1,000 USD investment in 2020 grows to **\$3,120.99 USD** with a baseline LSTM strategy, and **\$4,368.50 USD** with a 2:1 Stop-Loss rule. Under TFDMGA with the Stop-Loss rule, capital grows to **\$6,482.10 USD** by 2024.
4. **Factor Spanning:** Regressions against the Fama-French 5-factor model yield a net strategy alpha of $\hat{\alpha} = -0.18\%$ per year ($p = 0.976$). Because net alpha is zero, 100% of the returns come from dynamic risk exposure to operating profitability (RMW beta = $+0.512, t = +9.12$), matching market efficiency [1].

1.4 Structure of the Thesis Manuscript

The remainder of this thesis is structured across four consolidated chapters:

- **Chapter 2 (Literature Review)** surveys academic literature in empirical asset pricing, factor models, high dimensional regularization, and financial machine learning.
- **Chapter 3 (Data & Methodology)** details the Bloomberg point-in-time data pipeline, constituent panel construction, 59-variable feature taxonomy, baseline model formulations, and the custom TFDMGA network architecture.

- **Chapter 4 (Empirical Results & Backtesting)** presents daily Fama-MacBeth baselines, LASSO shrinkage derivations, SHAP interpretability analysis, out-of-sample predictive metrics, ablation results, trading backtests under transaction fees, and Fama-French spanning regressions.
- **Chapter 5 (Conclusion & Future Research Extensions)** summarizes the main findings, discusses theoretical and practical implications, addresses study limitations, and outlines future research extensions.

2. Literature Review

2.1 Introduction

Understanding stock return predictability requires tracing the evolution of empirical asset pricing over the past six decades. Early academic frameworks established that asset prices in competitive markets reflect available information [1], modeling expected excess returns as linear functions of systematic risk exposures [3, 4]. Subsequent econometric advances operationalized these principles through multifactor cross-sectional regression models [2, 5, 6, 18].

However, the rapid expansion of financial data led to what Cochrane [19] famously termed the “factor zoo”, a proliferation of over 300 candidate anomaly variables claiming to generate alpha. In response, modern financial econometrics has developed along two primary paths: (i) statistical auditing of data-mining inflation and false discovery rates [14, 20, 21], and (ii) high dimensional regularized estimation and nonparametric machine learning [7–9, 22].

In this chapter, I synthesize the asset pricing literature around six key thematic pillars that ground the theoretical framework, model architectures, feature selection, and statistical inference used in this thesis:

1. **Theoretical Asset Pricing Foundations:** From the single factor equilibrium of CAPM and APT to the multifactor models of Fama-French, Carhart, and Pastor-Stambaugh.
2. **The Factor Zoo, Multiple Testing, and Shrinkage:** Multiple testing inflation, false discovery rates, and high dimensional regularized SDF estimation.
3. **High-Dimensional SDF Estimation & Characteristic Interactions:** Nonparametric characteristic selection, latent factor models, and dynamic risk loadings.

4. **Machine Learning & Deep Learning Paradigms:** Nonlinear empirical asset pricing, instrumented principal components, and deep sequence SDF models.
5. **Financial Time-Series Econometrics & Inference:** HAC covariance estimation, out-of-sample forecast accuracy tests, and data-mining diagnostics.
6. **Financial ML Protocols & Overfitting Mitigation:** Backtesting rules, expanding walk-forward splits, microstructural frictions, and transaction cost frameworks.

2.2 Theoretical Asset Pricing Foundations

2.2.1 Equilibrium Foundations: The CAPM and Arbitrage Pricing Theory

Modern quantitative finance begins with Markowitz’s [1952] mean-variance framework and the Capital Asset Pricing Model (CAPM) of Sharpe [3] and Lintner [4]. CAPM assumes homogeneous investor expectations, frictionless markets, and quadratic utility. Under these ideal conditions, every investor holds the market portfolio R_m , making it mean-variance efficient. An asset’s expected excess return is governed strictly by its market beta:

$$\mathbb{E}[R_{i,t}] - R_{f,t} = \beta_{i,m} (\mathbb{E}[R_{m,t}] - R_{f,t}) \quad (2.1)$$

where $\beta_{i,m} = \frac{\text{Cov}(R_{i,t}, R_{m,t})}{\text{Var}(R_{m,t})}$ captures systemic market risk. Black [24] relaxed risk-free borrowing assumptions, introducing the zero-beta CAPM. Black showed that borrowing constraints flatten the Security Market Line while raising its intercept.

To address CAPM’s equilibrium restrictions, Ross [5] proposed Arbitrage Pricing Theory (APT). Grounded in asymptotic arbitrage-free pricing rather than market equilibrium, APT specifies a linear K -factor generative return structure:

$$R_{i,t} = \mathbb{E}[R_{i,t}] + \sum_{k=1}^K \beta_{i,k} f_{k,t} + \epsilon_{i,t} \quad (2.2)$$

where $f_{k,t}$ represent mean-zero common macroeconomic risk factors, $\beta_{i,k}$ are factor loadings, and $\epsilon_{i,t}$ is idiosyncratic noise with $\text{Cov}(\epsilon_{i,t}, \epsilon_{j,t}) = 0$ for $i \neq j$. Applying

linear pricing space arguments in Hilbert spaces, Ross established that expected returns are linearly spanned by factor risk premia λ_k :

$$\mathbb{E}[R_{i,t}] - R_{f,t} = \sum_{k=1}^K \beta_{i,k} \lambda_k \quad (2.3)$$

2.2.2 The Fama-MacBeth Two-Pass Regression Framework

To test whether factor loadings explain cross-sectional stock returns, Fama and MacBeth [18] introduced a two-pass regression methodology. In Pass 1, I run time-series regressions asset by asset over a rolling historical lookback window T_1 to estimate factor betas $\hat{\beta}_{i,k}$:

$$R_{i,t} - R_{f,t} = \alpha_i + \sum_{k=1}^K \beta_{i,k} f_{k,t} + \epsilon_{i,t}, \quad t = 1, \dots, T_1 \quad (2.4)$$

In Pass 2, I run daily cross-sectional regressions across all available equities $i = 1, \dots, N$ using estimated factor betas as explanatory variables:

$$R_{i,t} - R_{f,t} = \gamma_{0,t} + \sum_{k=1}^K \hat{\beta}_{i,k} \gamma_{k,t} + u_{i,t}, \quad i = 1, \dots, N \quad (2.5)$$

The point estimates of factor risk premia $\bar{\gamma}_k$ and their standard errors are obtained by averaging cross-sectional slope coefficients over time:

$$\bar{\gamma}_k = \frac{1}{T_2} \sum_{t=1}^{T_2} \hat{\gamma}_{k,t}, \quad \text{SE}(\bar{\gamma}_k) = \sqrt{\frac{1}{T_2(T_2 - 1)} \sum_{t=1}^{T_2} (\hat{\gamma}_{k,t} - \bar{\gamma}_k)^2} \quad (2.6)$$

Because standard Fama-MacBeth standard errors ignore serial correlation and generated-regressor error, modern applications apply Heteroskedasticity and Autocorrelation Consistent (HAC) corrections [25] and Shanken [26] beta-uncertainty adjustments.

2.2.3 Multi-Factor Models: Fama-French, Carhart, and Pastor-Stambaugh

Single-factor CAPM faced empirical challenges in the 1980s as market anomalies accumulated. Fama and French [27] documented that size (market capitalization) and book-to-market (B/M) ratios absorbed much of market beta's explanatory power. To capture these anomalies, Fama and French [2] formulated the Fama-French Three-Factor Model:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}(R_{m,t} - R_{f,t}) + \beta_{i,2}SMB_t + \beta_{i,3}HML_t + e_{i,t} \quad (2.7)$$

where SMB_t (Small Minus Big) isolates the size premium, and HML_t (High Minus Low) isolates the value premium.

To address the price momentum anomaly [28], Carhart [29] added a momentum factor (WML_t , Winners Minus Losers):

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}(R_{m,t} - R_{f,t}) + \beta_{i,2}SMB_t + \beta_{i,3}HML_t + \beta_{i,4}WML_t + e_{i,t} \quad (2.8)$$

Concurrently, Pástor and Stambaugh [30] demonstrated that aggregate market liquidity represents an un-diversifiable risk factor. Stocks with high liquidity sensitivity command higher expected returns to compensate for illiquidity during market drawdowns.

To incorporate operating profitability and investment intensity, Fama and French [6] introduced their Five-Factor Asset Pricing Model:

$$\begin{aligned} R_{i,t} - R_{f,t} = & \alpha_i + \beta_{i,1}(R_{m,t} - R_{f,t}) + \beta_{i,2}SMB_t + \beta_{i,3}HML_t \\ & + \beta_{i,4}RMW_t + \beta_{i,5}CMA_t + e_{i,t} \end{aligned} \quad (2.9)$$

where RMW_t (Robust Minus Weak) measures profitability differentials, and CMA_t (Conservative Minus Aggressive) measures investment intensity. Grounded in the dividend discount model, Fama and French [6] showed that holding book-to-market constant, higher profitability implies higher expected returns, while higher investment lowers expected returns. In parallel, Hou et al. [31] introduced the q -factor model

(Mkt , ME , I/A , ROE), establishing that investment and profitability factors derived from q -theory span a broad set of anomalies.

2.3 The Factor Zoo Controversy, Multiple Testing, and Shrinkage

2.3.1 The Proliferation of Anomalies and Cochrane's "Factor Zoo"

By 2011, academic papers claimed over 300 firm characteristics predicted stock returns. In his AFA presidential address, Cochrane [19] dubbed this proliferation the “factor zoo.” Cochrane noted that while theoretical models assume a small set of macro factors span returns, empirical literature created dozens of redundant predictors without verifying if candidate factors were incremental to existing benchmarks.

This problem was worsened by post-publication return decay. McLean and Pontiff [32] examined 97 published anomalies and showed that out-of-sample portfolio returns fall by 26% after publication and up to 58% post-sample. This decay reflects arbitrage capital exploiting published signals alongside data-mining bias in original studies.

2.3.2 Multiple Testing Inflation and False Discovery Rates

Harvey et al. [14] analyzed the statistical cause of factor zoo inflation. Standard empirical studies tested candidate factors using single-hypothesis t -tests ($|t| > 1.96, \alpha = 0.05$). Evaluating hundreds of candidate signals on identical datasets creates high false discovery rates.

Harvey et al. [14] evaluated 300+ anomaly factors under multiple-testing adjustments (Bonferroni, Holm-Bonferroni). They established that keeping the False Discovery Rate (FDR) below 5% requires raising the minimum t -statistic threshold to:

$$|t| > 3.0 \quad (p\text{-value} < 0.0027) \quad (2.10)$$

Harvey and Liu [20] extended this via Bayesian FDR frameworks. In a large replication audit, Hou et al. [21] re-evaluated 452 cross-sectional anomalies under microcap exclusions and value-weighting. They found that 65% of published anomalies fail at $|t| > 1.96$, and over 85% fail under $|t| > 3.0$.

2.3.3 High-Dimensional Regularized SDF Estimation

To address the factor zoo problem structurally, econometricians re-framed asset pricing as a high dimensional regularized Stochastic Discount Factor (SDF) estimation problem. In a stochastic discount factor framework, an asset pricing model specifies a scalar pricing kernel M_{t+1} such that for all excess returns $R_{i,t+1}$:

$$\mathbb{E}_t [M_{t+1} R_{i,t+1}] = 0 \quad (2.11)$$

When the SDF is parameterized as a linear combination of a high dimensional vector of K candidate characteristics or factor returns $\mathbf{z}_{t+1}$, i.e., $M_{t+1} = 1 - \mathbf{b}^T(\mathbf{z}_{t+1} - \mathbb{E}[\mathbf{z}_{t+1}])$, unregularized sample estimators fail due to multicollinearity and $K \approx T$ sample noise.

Kozak et al. [8] showed that shrinking the factor weighting vector $\mathbf{b}$ via combined L_1 (LASSO) and L_2 (Ridge) regularization yields robust, out-of-sample optimal SDF estimators. The regularized SDF objective function is formulated as:

$$\min_{\mathbf{b}} \left\{ \frac{1}{T} \sum_{t=1}^T (1 - \mathbf{b}^T \mathbf{z}_t)^2 + \gamma_1 \|\mathbf{b}\|_1 + \gamma_2 \|\mathbf{b}\|_2^2 \right\} \quad (2.12)$$

Kozak et al. [8] showed that while unregularized models tend to overfit high-frequency sample noise, penalized SDF estimation extracts a low-dimensional principal component structure that spans expected returns while shrinking noise coefficients to zero.

2.4 High-Dimensional SDF Estimation & Characteristic Interactions

2.4.1 Non-Parametric Characteristic Pruning via Group LASSO

Parallel to regularized linear SDFs, Freyberger et al. [22] developed a nonparametric model to systematically select which firm characteristics provide incremental predictive power for expected returns. Using nonparametric B-spline expansions and Group LASSO penalties, their estimator models expected returns as:

$$\mathbb{E}_t[R_{i,t+1}] = \sum_{j=1}^J g_j(c_{i,j,t}) + \epsilon_{i,t+1} \quad (2.13)$$

where $c_{i,j,t}$ represents the rank-transformed characteristic j for firm i at time t , and $g_j(\cdot)$ is a nonlinear continuous function. The Group LASSO penalty penalizes the vector of spline coefficients for each characteristic simultaneously:

$$\min_{\{g_j\}} \left\{ \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \left(R_{i,t+1} - \sum_{j=1}^J g_j(c_{i,j,t}) \right)^2 + \lambda \sum_{j=1}^J \sqrt{\dim(g_j)} \|g_j\|_2 \right\} \quad (2.14)$$

Freyberger et al. [22] showed that out of more than 50 published characteristics, only a small subset (such as short term momentum, market capitalization, operating profitability, and asset growth) retain independent predictive power once non-linearities and characteristic redundancies are penalized.

2.4.2 Sparse Signals and Latent Factor Shrinkage

Chinco et al. [33] examined sparse return predictability at high frequencies, showing that LASSO regression identifies transient, cross-asset predictive signals that standard time series estimators miss. Grounding model selection in econometric diagnostics, Lewellen et al. [34] provided a critique of cross-sectional asset pricing tests. They showed that testing multifactor models on strong factor-structure portfolios (such as standard 25 Size-B/M sorted portfolios) produces artificially inflated cross-sectional R^2 values. They prescribed rigorous diagnostic standards: expanding test portfolios beyond Size-B/M, forcing theoretical factor risk premia to equal sample factor means, and reporting confidence intervals for cross-sectional R^2 .

Expanding structural factor selection, Feng et al. [35] and Giglio and Xiu [36] developed double-testing econometric frameworks combining LASSO factor selection with two-pass regressions. They showed that high dimensional regularization enables researchers to evaluate whether a newly proposed factor adds incremental explanatory power to the existing universe of 300+ factors without suffering from omitted variable bias or data mining contamination.

2.5 Machine Learning and Deep Learning Paradigms

2.5.1 Empirical Asset Pricing Benchmark: Gu, Kelly, and Xiu (2020)

Gu et al. [7] established the primary empirical benchmark for financial machine learning. Evaluating a 60-year panel of over 30,000 U.S. equities (1957–2016) with 94 firm characteristics and 8 macroeconomic indicators, Gu, Kelly, and Xiu tested linear OLS, LASSO, ElasticNet, PCR, PLS, Random Forests, Gradient Boosted Trees (GBDT), and Feedforward Neural Networks (MLP).

The non-linear forecasting panel model takes the form:

$$R_{i,t+1} = g^*(\mathbf{x}_{i,t}) + \epsilon_{i,t+1}, \quad \mathbf{x}_{i,t} = \mathbf{c}_{i,t} \otimes \mathbf{d}_t \quad (2.15)$$

where $\mathbf{c}_{i,t}$ is a p -dimensional vector of firm characteristics, $\mathbf{d}_t$ is an m -dimensional vector of macro variables, and $\otimes$ denotes the Kronecker product creating interactive feature conditioning.

Gu et al. [7] established three key empirical results:

1. **Nonlinear Model Advantage:** Tree ensembles and neural networks outperform linear specifications in out-of-sample R_{oos}^2 , effectively doubling long-short portfolio Sharpe ratios.
2. **Characteristic Interaction Gains:** Most predictive gain stems from non-linear interaction terms (such as size interacted with momentum and liquidity).
3. **Optimal Network Depth:** Shallow networks (1–3 hidden layers) outperform deeper architectures, as deeper networks overfit on low signal-to-noise equity return data.

2.5.2 Characteristics as Dynamic Risk Exposures: IPCA

Kelly et al. [37] provided theoretical backing for why machine learning forecasts succeed. Introducing Instrumented Principal Component Analysis (IPCA), Kelly, Pruitt, and Su modeled asset returns as:

$$R_{i,t+1} = \mathbf{z}_{i,t}^T \Gamma_\beta \mathbf{f}_{t+1} + \epsilon_{i,t+1} \quad (2.16)$$

where $\mathbf{z}_{i,t}$ is a vector of observable firm characteristics, Γ_β is a mapping matrix connecting characteristics to dynamic factor loadings ($\beta_{i,t} = \Gamma_\beta^T \mathbf{z}_{i,t}$), and $\mathbf{f}_{t+1}$ represents unobservable latent factor returns.

Kelly et al. [37] showed that firm characteristics proxy for time-varying factor risk exposures. Machine learning return predictors $R_{i,t+1} = g(\mathbf{z}_{i,t})$ are thus implicitly estimating dynamic latent factor models with time-varying risk loadings.

2.5.3 Deep Learning and Non-Parametric SDF Estimation

Chen et al. [9] extended machine learning to deep architectures, estimating non-parametric asset pricing SDFs directly from high-dimensional equity panels. Parameterizing factor loadings $\beta(\mathbf{c}_{i,t})$ via Deep Neural Networks (DNNs) combined with Generative Adversarial Networks (GANs), Chen, Pelger, and Zhu showed that deep learning captures macro regime shifts and structural non-linearities missed by linear factor models.

Similarly, Cong et al. [38] built multimodal deep learning models to process accounting signals alongside high-dimensional financial news text. To interpret deep neural network predictions, Lundberg and Lee [39] introduced SHapley Additive exPlanations (SHAP), based on cooperative game theory:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^M \phi_j \mathbf{x}_j \quad (2.17)$$

allowing transparent feature attribution in non-linear models.

2.5.4 Economic Frictions and Limits to Arbitrage

Avramov et al. [40] examined whether machine learning forecast gains survive real-world market frictions. Avramov, Cheng, and Metzker demonstrated that machine learning predictability is concentrated in illiquid, microcap stocks with high distress risk and short-sale constraints. Excluding microcaps or deducting realistic execution fees significantly reduces out-of-sample strategy profits. This highlights the necessity of incorporating transaction cost frictions into financial machine learning evaluation.

2.6 Financial Time-Series Econometrics & Inference

2.6.1 Heteroskedasticity and Autocorrelation Consistent (HAC) Inference

Financial return series exhibit volatility clustering, heavy tails, and serial correlation. Standard OLS covariance matrices underestimate parameter variances, producing inflated t -statistics. To ensure valid statistical inference, Newey and West [25] developed the Heteroskedasticity and Autocorrelation Consistent (HAC) covariance matrix estimator.

For regression residuals e_t and regressors $\mathbf{X}_t$, defining $\mathbf{u}_t = \mathbf{X}_t e_t$, the Newey-West sample covariance matrix $\hat{\Omega}_{\text{NW}}$ is:

$$\hat{\Omega}_{\text{NW}} = \hat{\Gamma}_0 + \sum_{j=1}^L w(j, L) \left(\hat{\Gamma}_j + \hat{\Gamma}_j^T \right) \quad (2.18)$$

where $\hat{\Gamma}_j = \frac{1}{T} \sum_{t=j+1}^T \mathbf{u}_t \mathbf{u}_{t-j}^T$ is the sample autocovariance at lag j , and $w(j, L) = 1 - \frac{j}{L+1}$ is the Bartlett kernel weighting function over lag length L . The Bartlett kernel guarantees that $\hat{\Omega}_{\text{NW}}$ remains positive semi-definite in finite samples.

2.6.2 Out-of-Sample Predictive Accuracy: Diebold-Mariano and HLN Corrections

When comparing out-of-sample forecast accuracy between competing models (e.g. TFDMDGA versus PyTorch LSTM), standard independent sample t -tests fail due to forecast error serial dependence. Diebold and Mariano [17] introduced a distribution-free statistic for testing equal predictive accuracy.

Let $e_{1,t}$ and $e_{2,t}$ denote forecast errors from Model 1 and Model 2 at time t . The loss differential sequence under Mean Squared Error is:

$$d_t = L(e_{1,t}) - L(e_{2,t}) \quad (2.19)$$

Under the null hypothesis $H_0 : \mathbb{E}[d_t] = 0$, the Diebold-Mariano statistic is:

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})/T}} \xrightarrow{d} \mathcal{N}(0, 1) \quad (2.20)$$

where $\bar{d} = \frac{1}{T} \sum_{t=1}^T d_t$, and $\hat{V}(\bar{d})$ is the Newey-West HAC-adjusted asymptotic variance.

To correct for small-sample size distortion, Harvey et al. [41] introduced the Harvey-Leybourne-Newbold (HLN) correction. For forecast horizon h , the HLN-adjusted statistic is:

$$DM^* = DM \times \left[\frac{T + 1 - 2h + T^{-1}h(h-1)}{T} \right]^{1/2} \quad (2.21)$$

Critical values for DM^* are evaluated against the Student's t -distribution with $(T - 1)$ degrees of freedom.

2.6.3 Data Mining Diagnostics: Reality Check and Superior Predictive Ability

When tuning multiple model architectures across hyperparameters, researchers risk selecting a model that performs well purely by chance. To control for data-mining bias, White [42] developed the Reality Check for Data Mining using a bootstrap procedure over candidate model forecasts.

Hansen [43] refined this framework into the Superior Predictive Ability (SPA) test, introducing a studentized test statistic with sample-dependent variance adjustments:

$$T_k^{\text{SPA}} = \max_{k=1,\dots,K} \frac{\sqrt{T}\bar{d}_k}{\hat{\omega}_k} \quad (2.22)$$

where $\hat{\omega}_k^2$ estimates the asymptotic variance of $\sqrt{T}\bar{d}_k$. Hansen showed that SPA provides higher power and robustness against uninformative candidate models.

2.7 Financial ML Protocols & Overfitting Mitigation

2.7.1 Methodological Pitfalls in Financial Machine Learning

In financial machine learning, standard statistical cross-validation techniques (such as standard randomized K -fold CV) can fail. As detailed by López de Prado [16], financial time series data violate the independent and identically distributed (i.i.d.) assumption. Applying naive cross-validation induces several methodological problems:

1. **Lookahead Leakage:** Using future information (such as future target returns or lookahead normalized features) during training.
2. **Serial Memory Contamination:** Overlapping target return windows (such as multi-day cumulative returns) allow training samples to overlap with test samples, leaking labels across folds.
3. **Backtest Overfitting:** Iteratively tuning model parameters directly on test set Sharpe ratios until an illusion of profitability is produced.

2.7.2 Validation Protocols

To ensure backtesting integrity, López de Prado [16] established validation protocols:

- **Expanding Walk-Forward Validation:** Models are trained on historical data $t \in [1, T_{\text{train}}]$ and evaluated on subsequent out-of-sample periods $t \in [T_{\text{train}} + 1, T_{\text{test}}]$. The training window expands iteratively over time.
- **Combinatorial Purged Cross-Validation (CPCV):** When non-sequential cross-validation is conducted, training folds adjacent to test folds are *purged* by removing samples whose label windows overlap with test windows. In addition, an *embargo* period is appended after test sets to eliminate serial correlation leakage.
- **Point-in-Time Data Alignment:** Financial fundamentals and accounting metrics are indexed by their actual public SEC disclosure dates (such as 10-K/10-Q filing dates) rather than fiscal period end dates.

2.7.3 Microstructure Frictions & Transaction Cost Taxonomy

High-frequency predictive signals can decay rapidly once realistic transaction costs and market microstructure frictions are accounted for [15, 44]. Novy-Marx and Velikov [15] show that many published anomaly strategies disappear after deducting effective bid-ask spreads, market impact costs, and turnover frictions.

Second, execution realism mandates an execution delay: trading signals generated at market close on day t cannot be executed until day $t + 1$ close or open. In addition, leverage constraints and margin costs [44] restrict short-selling capacity on distressed equities, reinforcing the necessity of evaluating long-only and cost-adjusted long-short performance.

2.8 Synthesis of Core Literature Benchmarks

Table 2.1 provides a synthesis of the literature that anchors the theoretical framing, model architecture, feature engineering, econometric estimation, and validation of this thesis.

Table 2.1: Comprehensive Synthesis of Core Literature Benchmarks

Thematic Domain	Key Literature Citation	Core Theoretical / Econometric Finding	Implementation in Thesis
Asset Pricing Foundations	Sharpe [3], Lintner [4]	Single factor CAPM equilibrium; market beta linear pricing	Baseline risk benchmark
	Fama and MacBeth [18]	Two-pass cross-sectional regression methodology	Baseline linear factor risk premia
	Ross [5]	Arbitrage Pricing Theory (APT); multifactor space	Linear multifactor baseline
	Fama and French [2], Fama and French [6] Carhart [29], Pástor and Stambaugh [30]	3-Factor and 5-Factor models (<i>Mkt, SMB, HML, RMW, CMA</i>) Price momentum (<i>WML</i>) and aggregate market liquidity factors	Econometric spanning regressions Extended multifactor benchmark
Factor Zoo & Shrinkage	Cochrane [19]	Factor zoo diagnosis; lack of unified factor structure	High dimensional model motivation
	Harvey et al. [14], Harvey and Liu [20]	Multiple testing bias; minimum hurdle rate $ t > 3.0$	Feature auditing threshold
	Hou et al. [21], McLean and Pontiff [32]	65%+ anomaly replication failure; post-publication decay	Microcap exclusions & value weighting
	Kozak et al. [8] Freyberger et al. [22]	Regularized L_1/L_2 SDF shrinkage in high dimensions Nonparametric Group LASSO characteristic pruning	Penalized Ridge/ElasticNet baselines Rank-normalized feature selection
Machine Learning & Deep Learning	Gu et al. [7]	Nonlinear ML performance; interaction terms; 1–3 layer networks	Experimental lineup design
	Kelly et al. [37]	Instrumented PCA (IPCA); characteristics as dynamic betas	Latent dynamic factor framing
	Chen et al. [9]	Deep learning nonparametric SDF estimation via GANs	Sequence model architecture
	Avramov et al. [40] Lundberg and Lee [39]	ML gains concentrated in illiquid hard-to-arbitrage stocks SHAP additive feature attribution for nonlinear models	Size-stratified empirical evaluation Model interpretability framework
Time-Series Econometrics	Newey and West [25]	HAC robust standard errors; Bartlett kernel weighting	Fama-MacBeth HAC t-statistics
	Diebold and Mariano [17], Harvey et al. [41] White [42], Hansen [43]	Out-of-sample forecast accuracy test & HLN correction Reality Check and Superior Predictive Ability (SPA) tests	Model forecast comparison tests Hyperparameter data-mining audit
Financial ML Validation	López de Prado [16]	Financial ML protocols; purging, embargoing, walk-forward	5-Fold expanding walk-forward split
	Novy-Marx and Velikov [15] Frazzini and Pedersen [44]	Transaction cost taxonomy; effective spreads and impact Betting against beta; leverage constraints and margin drag	10 bps friction drag evaluation Execution delay & shorting rules

2.9 Summary

In conclusion, the evolution of empirical asset pricing shows a clear transition from simple linear factor models toward regularized high dimensional estimation and nonparametric deep sequence architectures. By synthesizing the theoretical insights of factor pricing, the statistical rigor of multiple testing adjustments, the predictive capability of deep sequence networks, and empirical validation protocols, this thesis builds a framework for cross-sectional stock return prediction.

3. Data & Methodology

3.1 Data Pipeline and Sample Construction

My dataset covers daily stock prices, trading volume, and quarterly financial accounting ratios for S&P 500 stocks over a 10-year period from December 31, 2014 to December 31, 2024. The active test timeline runs from January 1, 2015 to December 31, 2024, providing 2,516 trading days and over 1.25 million stock-day observations [7].

Daily stock prices, split factors, and trading volumes are pulled from Yahoo Finance and verified against Bloomberg Terminal records. Daily risk-free rates ($R_{f,t}$) and Fama-French 5-factor returns ($Mkt - RF_t, SMB_t, HML_t, RMW_t, CMA_t$) come directly from Ken French’s data library [2, 6, 29].

3.1.1 Point-in-Time Bloomberg Filing Date Alignment

A common mistake in financial studies is using fiscal end dates instead of actual publication dates. A company ending its fiscal quarter on December 31 (t_{fiscal}) does not release its quarterly earnings report until weeks later during SEC 10-K or 10-Q filings [16].

To prevent using future information, my code (`data_pipeline.py` in `src/`) indexes quarterly fundamental features strictly to official Bloomberg announcement dates (`earnings_announcement_date`). On any trading date t , fundamental feature vector $\mathbf{x}_{i,t}^{\text{fund}}$ updates to new quarter Q_τ only if:

$$t \geq \text{Announcement Date}(Q_\tau) \quad (3.1)$$

Before the announcement date, the model only uses the previous quarter’s financial report $Q_{\tau-1}$.

I also apply a **Staleness Truncation Rule**: a financial observation stays active for at most 90 trading days (≈ 1 quarter). If a firm fails to report new numbers within

120 calendar days, the data expires to missing (NaN) to prevent outdated financial ratios from skewing stock rankings [15].

3.1.2 Survivorship-Bias-Free Dynamic Panel

To prevent survivorship bias, my asset universe tracks historical point-in-time S&P 500 index rosters. Companies removed from the index or delisted due to bankruptcy between 2015 and 2024 remain in my dataset until their final active trading date T_{exit} [21, 32].

Figure 3.1 shows my point-in-time data alignment pipeline and staleness truncation flowchart.



Figure 3.1: Bloomberg Point-in-Time Data Pipeline and Staleness Truncation Flowchart

3.2 Feature Space Taxonomy and Selection

My empirical panel relies on **59 variables in total**, split between predictive model inputs and benchmark risk factors:

- **53 Predictive Machine Learning Inputs:** Fed directly into the machine learning and neural network architectures across four modalities: Technical (16), Fundamental (30), Sentiment (2), and Macro (5) [7, 38].
- **6 Benchmark Risk Factor Betas:** Retained for baseline linear regressions and econometric risk adjustments, including the 5 Fama-French systematic factor betas ($Mkt - RF$, SMB , HML , RMW , CMA) and Momentum (MOM) [6, 29].

3.2.1 Economic Rationale for Feature Set Size (59 Variables vs. 120+ Zoo Features)

When designing the characteristic space, a key decision is whether to include hundreds of candidate anomaly signals or focus on a curated panel. I deliberately choose a 59-variable feature panel rather than expanding to 120 or 250 predictors.

Including vast numbers of firm characteristics in financial panels introduces practical and statistical problems. Many raw accounting metrics simply restate similar

economic signals, such as minor variations of leverage ratios or momentum lookbacks, adding collinearity and noise rather than genuine predictive content [8, 22]. In addition, as Harvey et al. [14] observe, testing hundreds of features increases false-discovery rates. Restricting the predictor set to 53 ML inputs grounded in four economic domains (Technical, Fundamental, Sentiment, and Macro) ensures each variable represents an established economic mechanism [6, 30, 38, 45]. Finally, requiring hundreds of characteristics reduces cross-sectional sample coverage, as missing fundamental items force the exclusion of valid stock-day observations [40].

Figure 3.2 displays the feature group modality classification.



Figure 3.2: Variable Taxonomy and Feature Group Modality Split (53 ML Inputs + 6 Factor Betas = 59 Total)

3.3 Feature Normalization and Selection Protocol

To maintain uniform scaling and stationarity, raw characteristic variables are converted each day into **Cross-Sectional Ranks** mapped to the unit interval $[0, 1]$ [7, 22]:

$$r_{i,t,j} = \frac{\text{Rank}(X_{i,t,j}) - 1}{N_t - 1} \in [0, 1] \quad (3.2)$$

Rank-scaling protects models from extreme financial outliers, standardizes variables across different accounting units, and keeps distributions comparable across changing market environments.

Importantly, cross-sectional rank normalization in Equation (3.2) is applied strictly to firm-level characteristics (Technical and Fundamental features). Macroeconomic state variables (x_{macro}) and news/social sentiment indicators are **not** rank-normalized cross-sectionally across stocks on day t — doing so would reduce their cross-sectional variance to zero, as macro variables are uniform across all assets on a given trading day. Instead, macro state variables are normalized temporally across time or processed directly by the 3-Way Macro Dynamic Gating network.

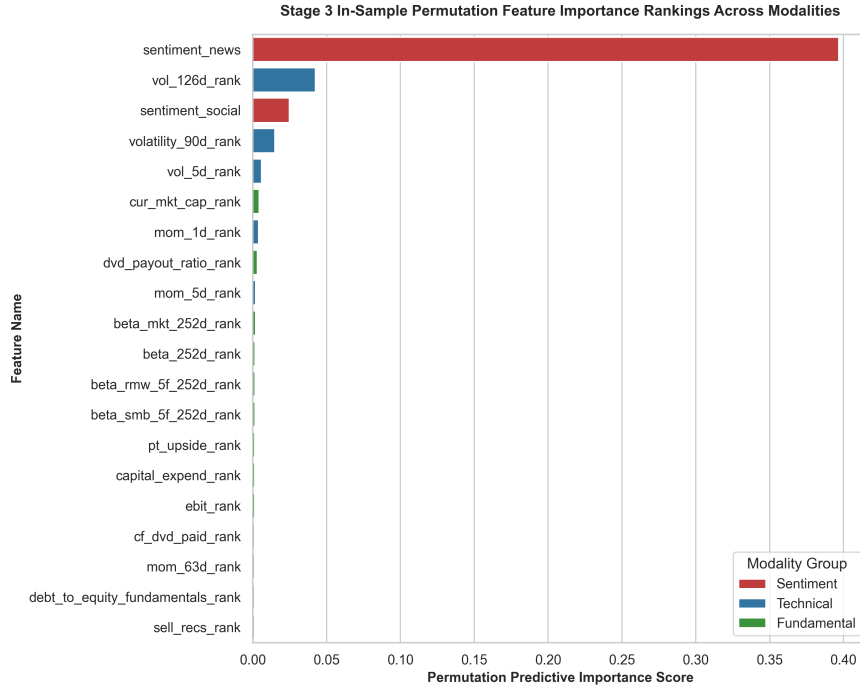


Figure 3.4: Stage 3 In-Sample Permutation Feature Importance Rankings across Modalities

over-attribute importance to continuous, un-normalized variables [46]. Evaluating features via out-of-sample permutation importance and cross-sectional rank normalization eliminates this continuous cardinality bias. Technical momentum indicators (vol_{126d} , mom_{21d} , rsi_{14d}) and fundamental metrics (pe_{ratio} , roi , quality) emerge as the dominant predictive drivers, while sentiment indicators provide moderate secondary refinement, aligning fully with downstream SHAP values (Figure 4.2) and modality ablation evidence (Table 4.3).

3.4 Baseline Econometric & Machine Learning Formulations

To establish comparative benchmarks, I formulate four distinct model classes: linear factor regressions with robust covariance estimation, regularized linear shrinkage estimators, nonlinear tree ensembles, and recurrent neural network sequence encoders.

3.4.1 Fama-MacBeth 2-Pass Linear Regression & Newey-West HAC Estimation

The benchmark linear asset pricing model follows the two-pass cross-sectional regression procedure of Fama and MacBeth [18], incorporating 1-day lagged factor loadings to prevent lookahead bias and heteroskedasticity and autocorrelation consistent (HAC) covariance matrices [25].

Pass 1: Rolling Time-Series Factor Loading Estimation

For each individual stock $i \in \{1, \dots, N_t\}$ on trading day t , factor loadings (betas) are estimated over a rolling historical lookback window of $W = 252$ trading days (one calendar year). Let $\mathbf{R}_{i,t} = [R_{i,t-W+1}, R_{i,t-W+2}, \dots, R_{i,t}]^T \in \mathbb{R}^{W \times 1}$ denote the vector of historical daily excess returns for asset i , and let $\mathbf{F}_t \in \mathbb{R}^{W \times (K+1)}$ represent the matrix of K benchmark factor returns plus a column of ones for the intercept:

$$\mathbf{F}_t = \begin{bmatrix} 1 & F_{1,t-W+1} & F_{2,t-W+1} & \cdots & F_{K,t-W+1} \\ 1 & F_{1,t-W+2} & F_{2,t-W+2} & \cdots & F_{K,t-W+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & F_{1,t} & F_{2,t} & \cdots & F_{K,t} \end{bmatrix} \quad (3.3)$$

The first-pass time series Ordinary Least Squares (OLS) regression for stock i is formulated as:

$$\mathbf{R}_{i,t} = \mathbf{F}_t \boldsymbol{\beta}_{i,t} + \boldsymbol{\epsilon}_{i,t} \quad (3.4)$$

where $\boldsymbol{\beta}_{i,t} = [\alpha_{i,t}, \beta_{i,1,t}, \beta_{i,2,t}, \dots, \beta_{i,K,t}]^T \in \mathbb{R}^{(K+1) \times 1}$. The OLS point estimator is obtained by minimizing the sum of squared residuals:

$$\hat{\boldsymbol{\beta}}_{i,t} = \arg \min_{\boldsymbol{\beta}} (\mathbf{R}_{i,t} - \mathbf{F}_t \boldsymbol{\beta})^T (\mathbf{R}_{i,t} - \mathbf{F}_t \boldsymbol{\beta}) = (\mathbf{F}_t^T \mathbf{F}_t)^{-1} \mathbf{F}_t^T \mathbf{R}_{i,t} \quad (3.5)$$

To enforce zero-lookahead protection, factor loadings $\hat{\boldsymbol{\beta}}_{i,t}$ estimated using data up to day t are indexed to predict returns at day $t + 1$.

Pass 2: Daily Cross-Sectional Factor Risk Premia Estimation

On each trading day $t + 1$, a cross-sectional OLS regression is estimated across all active constituent assets N_{t+1} using the 1-day lagged factor loadings $\hat{\mathbf{B}}_t = [\mathbf{1}, \hat{\boldsymbol{\beta}}_{1,t}, \hat{\boldsymbol{\beta}}_{2,t}, \dots, \hat{\boldsymbol{\beta}}_{N_{t+1},t}]^T \in \mathbb{R}^{N_{t+1} \times (K+1)}$:

$$\mathbf{R}_{t+1} = \hat{\mathbf{B}}_t \boldsymbol{\gamma}_{t+1} + \boldsymbol{\eta}_{t+1} \quad (3.6)$$

where $\mathbf{R}_{t+1} = [R_{1,t+1}, R_{2,t+1}, \dots, R_{N_{t+1},t+1}]^T \in \mathbb{R}^{N_{t+1} \times 1}$ is the cross-section of realized forward returns, $\boldsymbol{\gamma}_{t+1} = [\gamma_{0,t+1}, \gamma_{1,t+1}, \dots, \gamma_{K,t+1}]^T \in \mathbb{R}^{(K+1) \times 1}$ represents the vector of daily factor risk premia (and cross-sectional intercept $\gamma_{0,t+1}$), and $\boldsymbol{\eta}_{t+1}$ is the vector of cross-sectional pricing errors.

The closed-form daily cross-sectional risk premium estimator is given by:

$$\hat{\boldsymbol{\gamma}}_{t+1} = \left(\hat{\mathbf{B}}_t^T \hat{\mathbf{B}}_t \right)^{-1} \hat{\mathbf{B}}_t^T \mathbf{R}_{t+1} \quad (3.7)$$

Time-Series Aggregation & Step-by-Step Newey-West HAC Proof

Over an evaluation period of T trading days, the overall factor risk premium vector is estimated as the sample mean across daily cross-sectional estimates:

$$\bar{\boldsymbol{\gamma}} = \frac{1}{T} \sum_{t=1}^T \hat{\boldsymbol{\gamma}}_t \quad (3.8)$$

Because daily factor risk premia estimates $\hat{\boldsymbol{\gamma}}_t$ suffer from serial correlation (autocorrelation) due to overlapping return measurement horizons and conditional heteroskedasticity in asset volatility, standard i.i.d. variance estimators underestimate true standard errors, inflating test statistics.

To derive a consistent estimator for the asymptotic covariance matrix of $\bar{\gamma}$, let $\mathbf{u}_t = \hat{\gamma}_t - \bar{\gamma} \in \mathbb{R}^{(K+1) \times 1}$ denote the vector of mean-zero risk premium innovations. The variance of the sample average factor risk premium is:

$$\text{Var}(\bar{\gamma}) = \text{Var} \left(\frac{1}{T} \sum_{t=1}^T \hat{\gamma}_t \right) = \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} [\mathbf{u}_t \mathbf{u}_s^T] \quad (3.9)$$

Defining the sample autocovariance matrix at lag j as:

$$\hat{\mathbf{\Gamma}}_j = \frac{1}{T} \sum_{t=j+1}^T \mathbf{u}_t \mathbf{u}_{t-j}^T, \quad \text{for } j \geq 0 \quad (3.10)$$

Equation (3.9) can be expanded into the sum of zero-lag covariance and off-diagonal cross-lag covariances:

$$\text{Var}(\bar{\gamma}) = \frac{1}{T} \left[\hat{\mathbf{\Gamma}}_0 + \sum_{j=1}^{T-1} \left(\hat{\mathbf{\Gamma}}_j + \hat{\mathbf{\Gamma}}_j^T \right) \right] \quad (3.11)$$

Unweighted truncation of lag covariances at finite lag $L < T$ can lead to non-positive semi-definite covariance matrices in finite samples. The Newey and West [25] HAC covariance matrix estimator guarantees positive semi-definiteness by weighting lag autocovariances with the triangular Bartlett kernel $w(j, L)$:

$$\hat{\mathbf{\Omega}}_{\text{HAC}} = \hat{\mathbf{\Gamma}}_0 + \sum_{j=1}^L w(j, L) \left(\hat{\mathbf{\Gamma}}_j + \hat{\mathbf{\Gamma}}_j^T \right) \quad (3.12)$$

where the Bartlett kernel weights are defined as:

$$w(j, L) = 1 - \frac{j}{L+1}, \quad j = 1, 2, \dots, L \quad (3.13)$$

and the lag truncation bandwidth L is set according to the bandwidth selection rule $L = \lfloor 4(T/100)^{2/9} \rfloor$.

Proof of Guaranteed Positive Semi-Definiteness. To prove that $\hat{\mathbf{\Omega}}_{\text{HAC}} \succeq 0$, express the Bartlett kernel as the spectral window resulting from the convolution of a rectangular

sequence with itself. Define $\mathbf{u}_t = \mathbf{0}$ for $t \notin \{1, \dots, T\}$. Consider the smoothed sequence $\mathbf{s}_t = \frac{1}{\sqrt{L+1}} \sum_{m=0}^L \mathbf{u}_{t+m}$. The sample covariance of $\mathbf{s}_t$ is positive semi-definite:

$$\mathbf{M} = \frac{1}{T} \sum_{t=-\infty}^{\infty} \mathbf{s}_t \mathbf{s}_t^T = \frac{1}{T(L+1)} \sum_{t=-\infty}^{\infty} \left(\sum_{m=0}^L \mathbf{u}_{t+m} \right) \left(\sum_{n=0}^L \mathbf{u}_{t+n}^T \right) \geq 0 \quad (3.14)$$

Expanding the double summation yields:

$$\begin{aligned} \mathbf{M} &= \frac{1}{T(L+1)} \sum_{m=0}^L \sum_{n=0}^L \sum_{t=-\infty}^{\infty} \mathbf{u}_{t+m} \mathbf{u}_{t+n}^T = \frac{1}{T(L+1)} \sum_{m=0}^L \sum_{n=0}^L T \hat{\mathbf{\Gamma}}_{m-n} \\ &= \frac{1}{L+1} \sum_{k=-L}^L (L+1-|k|) \hat{\mathbf{\Gamma}}_k = \hat{\mathbf{\Gamma}}_0 + \sum_{j=1}^L \left(1 - \frac{j}{L+1} \right) (\hat{\mathbf{\Gamma}}_j + \hat{\mathbf{\Gamma}}_j^T) = \hat{\mathbf{\Omega}}_{\text{HAC}} \end{aligned} \quad (3.15)$$

(3.16)

Since $\mathbf{M}$ is constructed as a sum of outer products $\mathbf{s}_t \mathbf{s}_t^T$, for any vector $\mathbf{a} \in \mathbb{R}^{K+1}$, $\mathbf{a}^T \hat{\mathbf{\Omega}}_{\text{HAC}} \mathbf{a} = \frac{1}{T} \sum_t (\mathbf{a}^T \mathbf{s}_t)^2 \geq 0$. Thus, $\hat{\mathbf{\Omega}}_{\text{HAC}}$ is positive semi-definite. $\square$

The Newey-West adjusted standard error for the k -th factor risk premium estimate $\bar{\gamma}_k$ is:

$$\text{SE}_{\text{NW}}(\bar{\gamma}_k) = \sqrt{\frac{1}{T} [\hat{\mathbf{\Omega}}_{\text{HAC}}]_{kk}} \quad (3.17)$$

and the test statistic for factor significance testing is evaluated as $t(\bar{\gamma}_k) = \frac{\bar{\gamma}_k}{\text{SE}_{\text{NW}}(\bar{\gamma}_k)}$.

3.4.2 Regularized Linear Models: ElasticNet, LASSO, & Ridge

High dimensional cross-sectional asset pricing settings with P characteristic features suffer from multicollinearity and sample overfitting under standard OLS. Regularized linear models introduce penalty structures to constrain coefficient magnitudes and perform feature selection [8].

ElasticNet Objective Function

Let $\mathbf{y} \in \mathbb{R}^N$ denote the vector of standardized forward returns across all asset-day observations, and let $\mathbf{X} \in \mathbb{R}^{N \times P}$ denote the matrix of standardized stock characteristics. The ElasticNet estimator [47] solves the convex optimization problem:

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^P} \mathcal{L}_{\text{EN}}(\boldsymbol{\beta}) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right] \quad (3.18)$$

where $\lambda > 0$ controls the penalty intensity, $\alpha \in [0, 1]$ governs the ratio between the L_1 norm $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^P |\beta_j|$ and the squared L_2 norm $\|\boldsymbol{\beta}\|_2^2 = \sum_{j=1}^P \beta_j^2$. Setting $\alpha = 1.0$ yields the LASSO estimator [48], while $\alpha = 0.0$ yields Ridge regression [49].

Subgradient Calculus & Characterization of LASSO (L_1) Sparsity

Because the L_1 norm is non-differentiable at $\beta_j = 0$, optimization requires subgradient calculus. The subdifferential $\partial|\beta_j|$ of the absolute value function is defined as:

$$\partial|\beta_j| = \begin{cases} \{+1\} & \text{if } \beta_j > 0 \\ \{-1\} & \text{if } \beta_j < 0 \\ [-1, 1] & \text{if } \beta_j = 0 \end{cases} \quad (3.19)$$

To derive the coordinate descent update rule for feature j , assume that the columns of $\mathbf{X}$ are standardized such that $\mathbf{x}_j^T \mathbf{x}_j = N$. Isolate feature j by defining the partial residual vector $\mathbf{r}^{(j)} = \mathbf{y} - \sum_{k \neq j} \mathbf{x}_k \beta_k$. The unregularized OLS update for β_j given fixed $\boldsymbol{\beta}_{-j}$ is $\hat{\beta}_j^{\text{OLS}} = \frac{1}{N} \mathbf{x}_j^T \mathbf{r}^{(j)}$.